

1.2 Rational functions and graphs

Further Pure 1 · 2026 examination · Concept guide and original exam-style practice

Understand the concept

A rational graph is controlled by zeros of numerator, excluded denominator values, asymptotes and possible cancellations. Divide polynomials for an oblique asymptote. Differentiation locates stationary points; sign charts show intervals where the curve rises and falls. A cancelled factor may produce a hole rather than a vertical asymptote.

Worked example: follow the reasoning

For $y = (x^2 - 1)/(x - 1)$, $x \neq 1$, the curve is $y = x + 1$ with a hole at $(1, 2)$.

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Find vertical and horizontal asymptotes of $y = (2x + 1)/(x - 3)$ and identify the x-intercept.

[3]

Final answer: _____

Answers and explanation

Question 1 [3 marks]

Denominator zero gives $x=3$ [1]; ratio leading coefficients $y=2$ [1]; numerator zero gives $x=-1/2$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Further Mathematics 9231, 2026–2027.
<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.